Statistical Case Studies - Project 1 (Part 1)

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Executive Summary

Current flood risk models generally assume that the shape of a flood is fixed, varying only in its maximum river flow. This project investigates whether this approach captures the uncertainty in flood events on the River Don at Parkhill, by considering both the maximum flow and the time a flood takes to reach its peak.

Using ten years of river discharge data, we identified 44 independent extreme flow events, across which both quantities varied considerably, with median values of around 106 m^3/s and 44 hours respectively. Flood events were concentrated in autumn and winter, with autumn events typically taking longer to reach their peak, and we found only weak evidence of a relationship between maximum flow and time to peak, suggesting that larger floods do not necessarily develop more slowly.

These findings suggest that assuming a fixed hydrograph shape may overlook important uncertainty, so SEPA should consider incorporating variation in time to peak alongside maximum flow in future flood risk modelling guidance.

Data and Exploratory Analysis

Data Description and Quality

Table 1: Data used in this analysis, and the SEPA quality codes assigned to each reading.

Data description	
Station	Don at Parkhill
Source	SEPA time series service (KiWIS)
Measurement	River discharge (m^3/s)
Resolution	15 minutes
Period	01 October 2016 – 30 September 2026
Readings	350,592
Missing readings	0
Quality codes	
50 – Good (G)	335,515 readings (95.70%), retained
100 – Estimated (E)	14,915 readings (4.25%), retained
140 – Provisional (Prov)	0 readings (0.00%), excluded
150 – Suspect (S)	0 readings (0.00%), excluded
200 – Unchecked (imported from legacy database) (V)	0 readings (0.00%), excluded
254 – Unchecked (U)	162 readings (0.05%), excluded

Unchecked readings (n = 162) occur from 29 September 2026, 16:15 onward and in short bursts of at most 45 minutes in 2017–2018.

The subsequent analysis is based on 350,592 discharge readings (m^3/s) from the River Don at Parkhill velocity-area gauging station, recorded every 15-minutes between 1 October 2016 and 30 September 2026, covering 10 water years. Parkhill is the lowest gauging station on the Don, with a catchment area of 1,273 km^2, and applies a single peak flow rating curve to all measurements.

The dataset is complete, and each reading carries a SEPA quality code (Table 1). We excluded the 162 Unchecked readings, since only two coincided with extreme flows, and retained the 14,915 Estimated readings, which extrapolate the rating curve above the 3.7 m gauging limit and therefore contain the largest recorded flows (Figure 1).

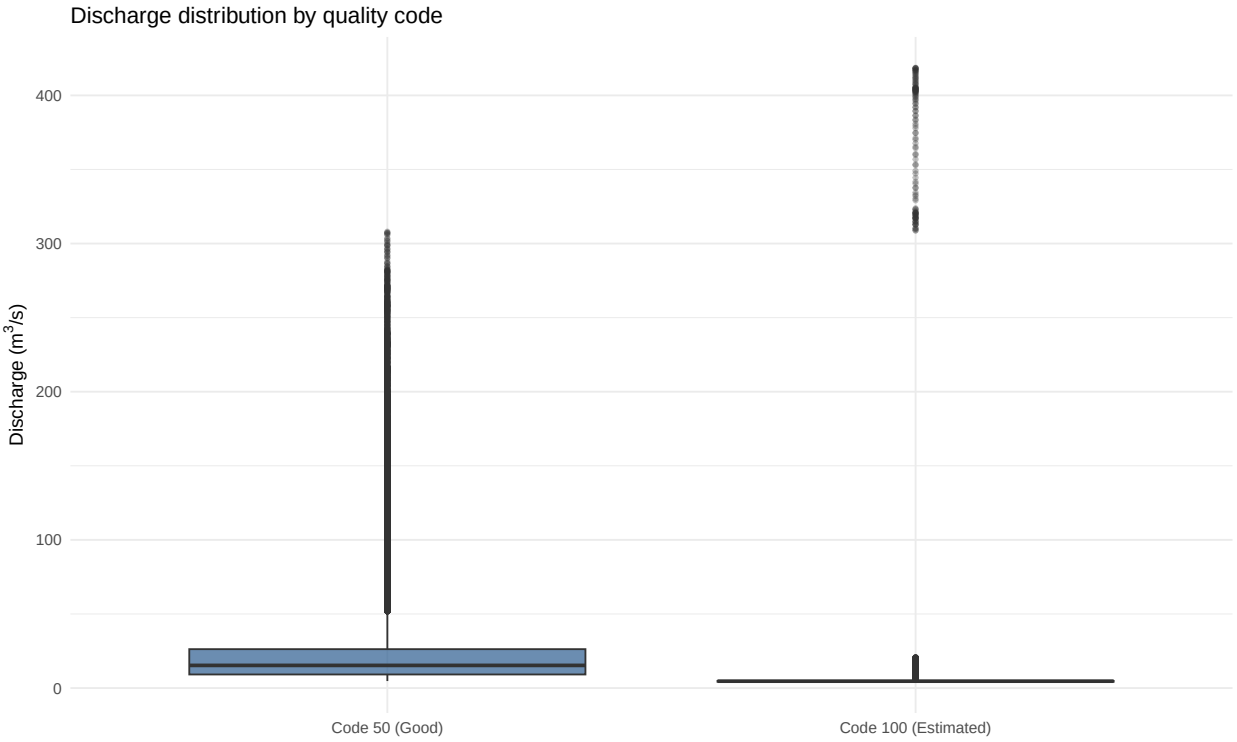


Figure 1: Discharge distribution for quality codes 50 & 100.

Exploratory Analysis

Hydrograph and Distribution

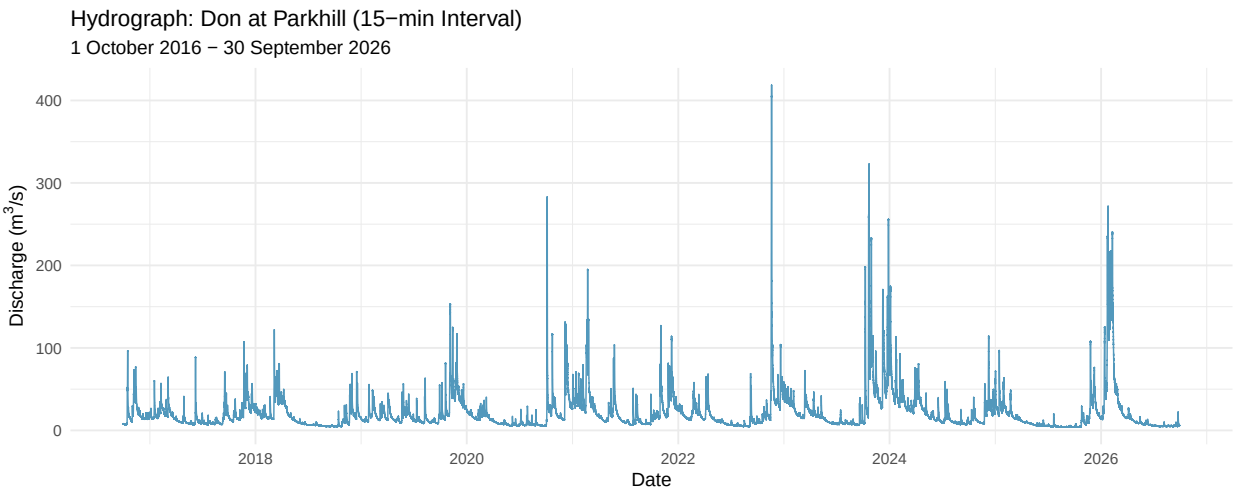


Figure 2: Hydrograph of 15-minute discharge for the Don at Parkhill.

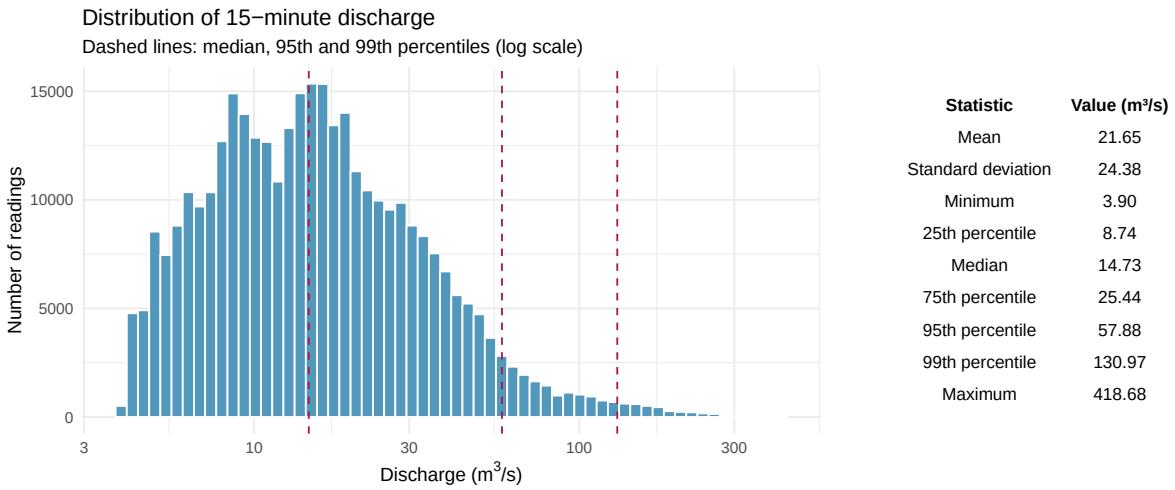


Figure 3: Distribution of 15-minute discharge on a log scale, with summary statistics. Dashed lines mark the median, 95th and 99th percentiles.

From Figure 2, we observe that the discharge is low for most of the year, with a smaller number of large peaks, the largest reaching 418.68 m^3/s on 19th November 2022. The distribution in Figure 3 is strongly right-skewed, with a median of 14.73 m^3/s compared to a 99th percentile of 130.97 m^3/s, which initialises our view that extreme flows behave very differently from typical flows and need to be analysed separately.

Seasonality of High Flows

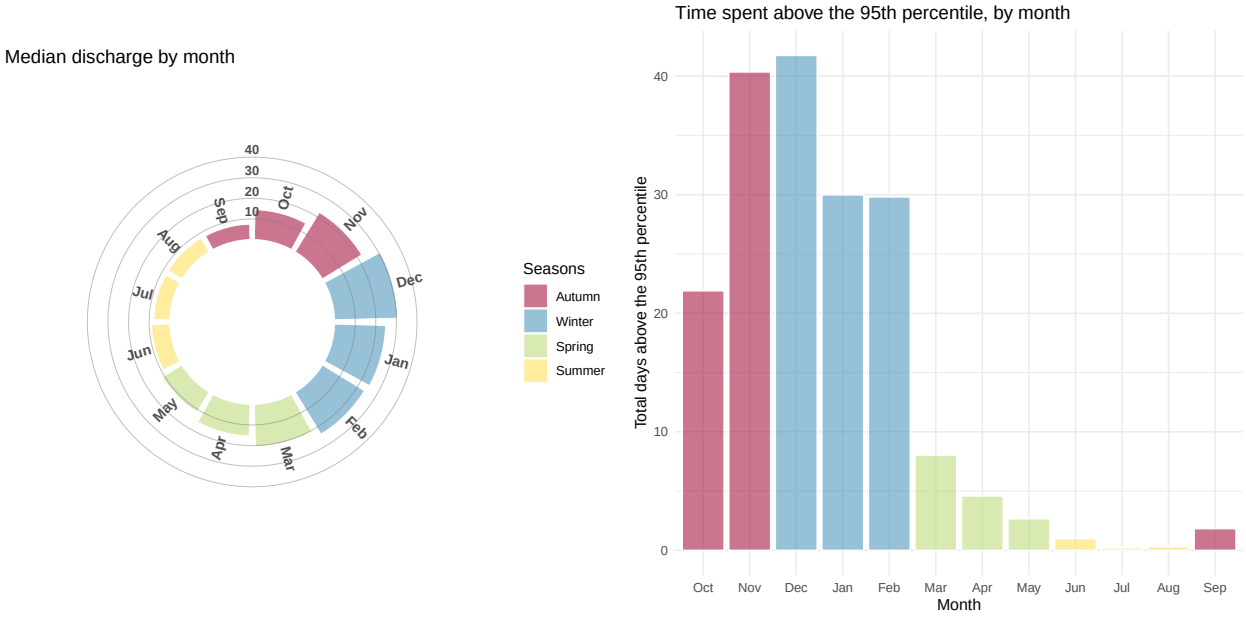


Figure 4: Median discharge by month (left) and total days spent above the 95th percentile in each month (right).

We observe that median discharge is highest from December to February and lowest from July to September, while time spent above the 95th percentile is even more concentrated to between October and February.

Flood Event Hydrographs

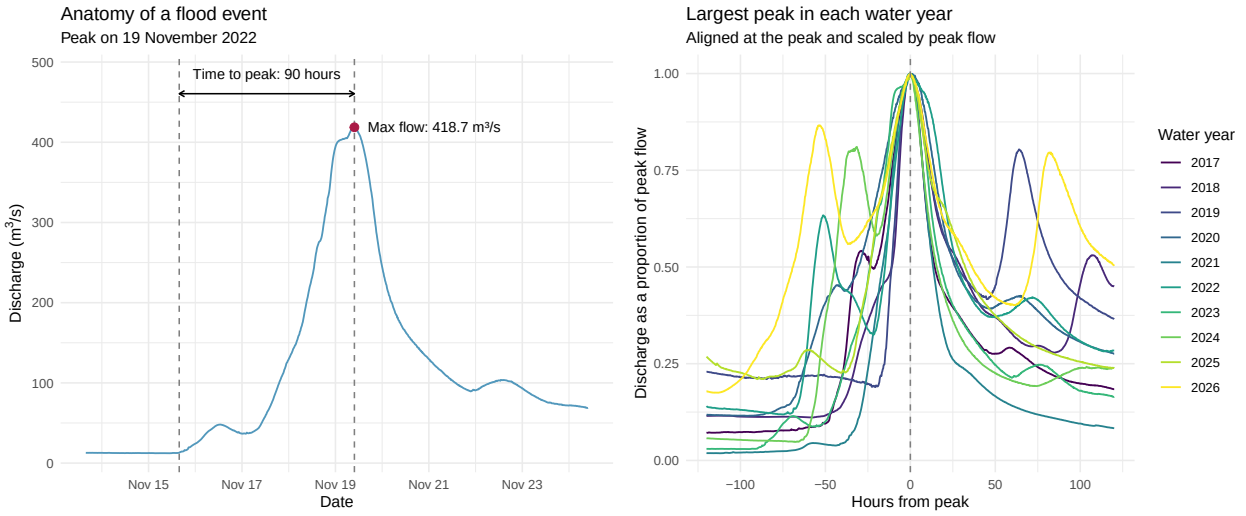


Figure 5: The largest recorded flood event, showing the max flow and time to peak (left), and the largest peak in each water year, aligned at the peak and scaled by peak flow (right).

This introduces a key assumption that we will use throughout the report:

- Start of the rise: the last reading before the peak with flow at most $q_{\min} + 0.05\,(q_{\text{peak}} - q_{\min})$, where $q_{\min}$ is the lowest flow in the previous 7 days.

The left panel of Figure 4 shows the two quantities for the largest observed event, and in the right panel we observe that the largest peak in each water year varies considerably in shape, which suggests that the current approaches to flood inundation modelling of treating the hydrograph shape as fixed ignores an important source of uncertainty. Additionally, we observe the number of peaks clustered around each largest peak, highlighting the importance of considering independent events.

Summary

Since high flows are rare, concentrated between October and February, and vary in shape between events, we need a threshold that extracts them. We will then separate peaks into independent events, before investigating how max flow and time to peak vary across events and between seasons.

Threshold Selection and Independence

Initial Threshold

The Peaks Over Threshold (POT) approach extracts flood events as the peaks of flow above a threshold u. Two common rules for choosing u are the 95th percentile of flow and a threshold giving 3 to 5 peaks per year. Figure 6 shows both on the hydrograph, where the 95th percentile is 57.9 m³/s, and counting each period above the threshold as a peak gives at least 5, 4 and 3 peaks per year at 79, 93 and 113 m³/s. At this stage, however, a single flood can cross the threshold several times, so these counts overstate the number of distinct events. Robson and Reed (1999) note that a threshold giving around five peaks a year leaves the flexibility to raise it later to exclude smaller floods, and the lecture notes that the threshold may need to change once independence has been applied. Since the 95th percentile gives 9.4, well above this range, we take the threshold giving 5 peaks per year $u = 79\ m^3/s$, as our initial threshold.

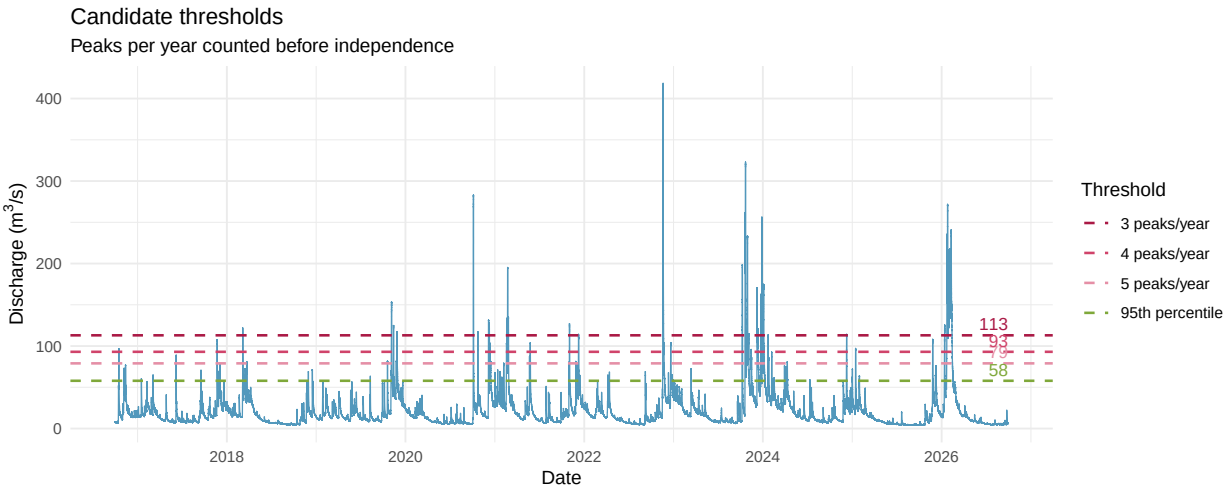


Figure 6: Hydrograph with the candidate thresholds. Peaks per year are counted before applying the independence rules.

Independence of Peaks

To make sure each peak represents a separate flood, we apply the independence rules of Robson and Reed (1999). Two adjacent peaks are only treated as separate events if:

1. they are at least three times the average time to rise apart, where the time to rise is the time between a trough and its peak, averaged over at least five clean peaks above the threshold; and
2. the flow at the trough between them is at least two-thirds lower than the first peak, i.e. below a third of it.

If either rule fails, we treat the peaks as one event and keep the larger peak. In applying these rules, we use the following assumptions:

- Clean peak: a peak with no other candidate peak in the previous 7 days.
- Trough: the start of the rise defined in the exploratory analysis, which we use to calculate the time to rise.

At the initial threshold, the average time to rise is 57.2 hours, so peaks must be at least 171.6 hours apart. This reduces the 56 candidate peaks to 35 independent events, or 3.5 per year (Figure 7).

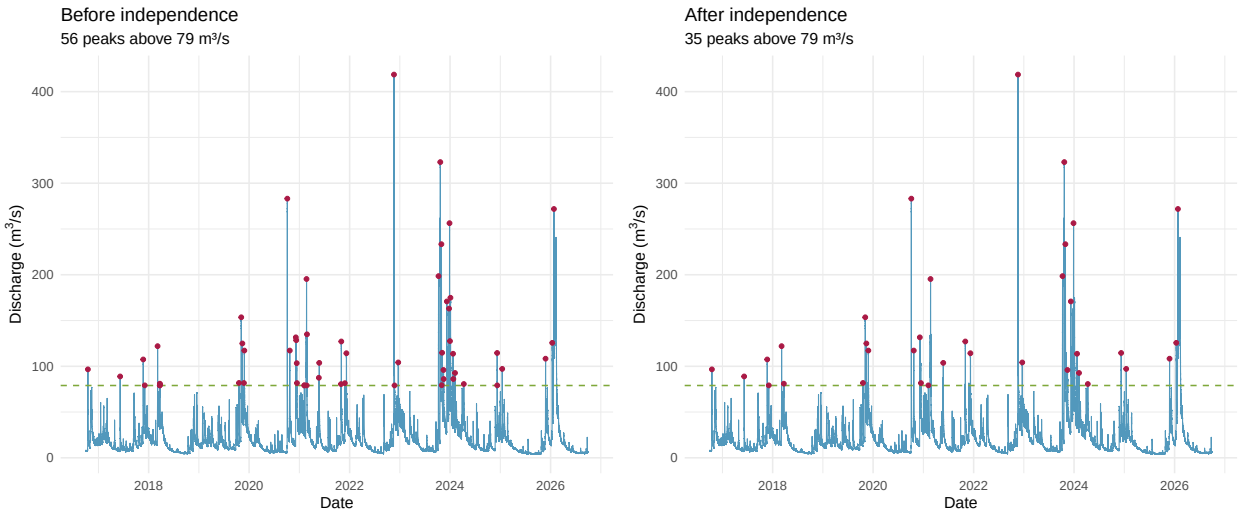


Figure 7: Peaks above the initial threshold before (left) and after (right) applying the independence rules.

Final Threshold

Since separating events lowers the number of peaks per year, we recheck the threshold using independent peaks only. Our initial threshold gives 3.5 independent peaks per year, while Figure 8 shows that thresholds between 57 and 81 m^3/s give 3 to 5.

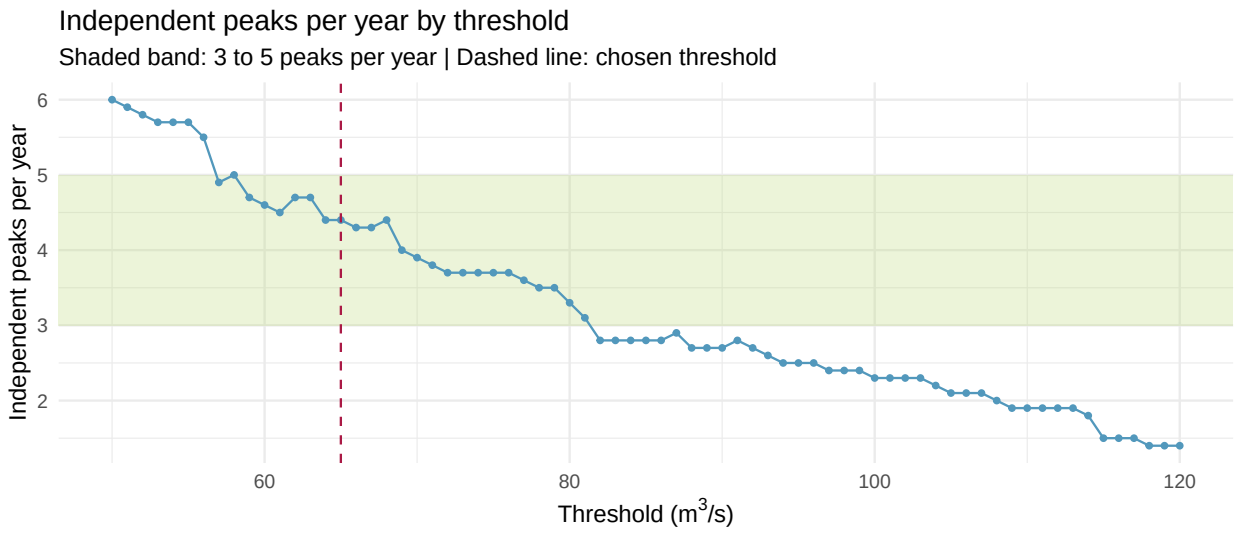


Figure 8: Independent peaks per year at each threshold.

To check which of these is suitable for extreme value modelling, we fit a Generalised Pareto Distribution (GPD) to the excesses of the independent peak flows X above u,

$$P(X - u \leq y \mid X > u) = 1 - \left(1 + \frac{\xi y}{\sigma}\right)^{-1/\xi}, \quad y > 0, \quad 1 + \frac{\xi y}{\sigma} > 0,$$

where $\sigma > 0$ is the scale and $\xi \neq 0$ the shape parameter, with $\xi = 0$ giving an exponential tail as a limiting case. If the GPD holds above u, the mean excess $E[X - u \mid X > u] = (\sigma_0 + \xi u)/(1 - \xi)$ is linear in u (for $\xi < 1$), where σ_0 is the scale parameter at a threshold of 0, and the estimates of ξ and the modified scale $\sigma - \xi u$ stay roughly constant as the threshold increases (Coles, 2001). [From Figure X, the mean residual life plot is approximately linear above … m³/s, and the parameter estimates are stable from … m³/s.]

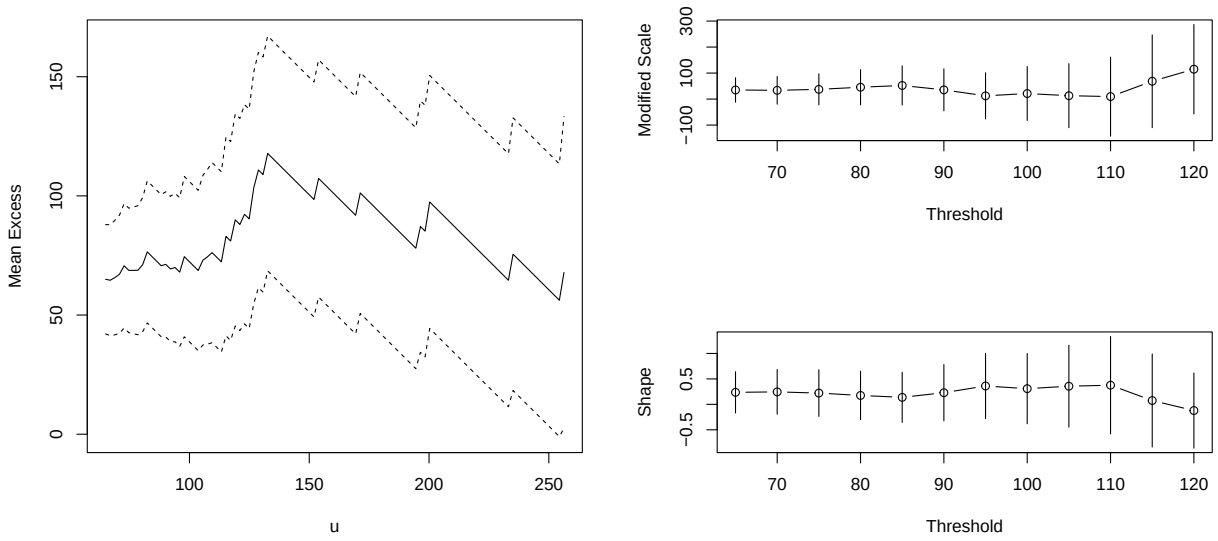


Figure 9: Mean residual life plot (left) and parameter stability plots (right) for the independent peaks at the final threshold.

Since the lecture notes that the threshold may need to change once independence has been applied, we lower it to a final threshold of $u = 65$ m^3/s, [which ...], giving 44 independent events (4.4 per year), with fitted parameters $\hat{\sigma} = 50.34$ (standard error 12.76) and $\hat{\xi} = 0.236$ (standard error 0.207). [The diagnostic plots in Figure 9 show ...]

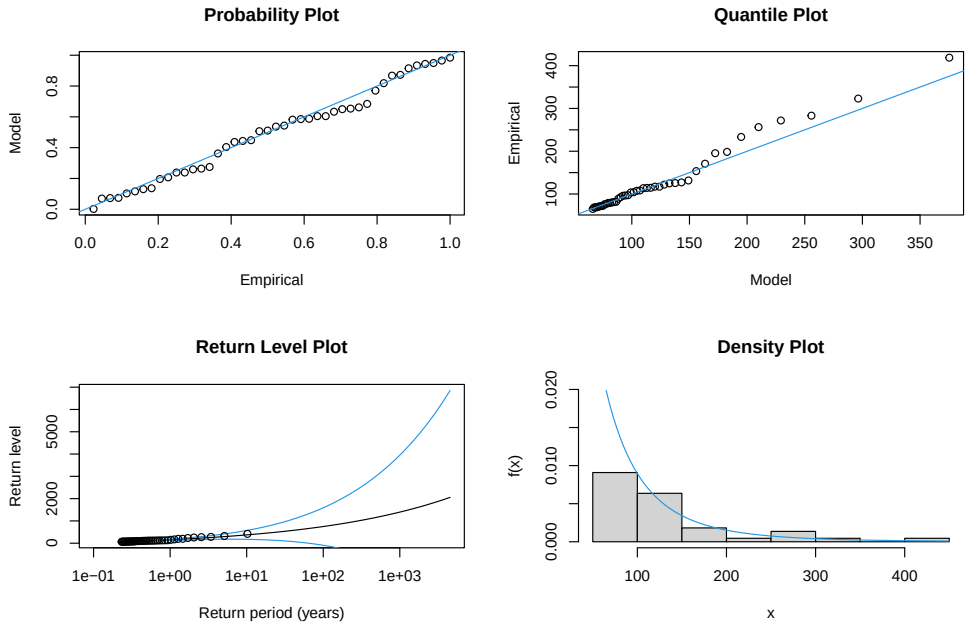


Figure 10: Diagnostic plots for the GPD fitted at the final threshold.

Uncertainty Analysis

The final threshold of $u = 65$m^3/s, identifies 44 independent flood events over the ten water years. Throughout this period, maximum flow and time to peak varied considerably between events, with medians of 105.83m^3/s and 44.13 hours respectively. The distributions and scatterplot (Figure 11, 12) illustrate variability in both quantities, suggesting that extreme flood hyrdographs cannot be adequately characterised by a single peak magnitude or a fixed time to peak.

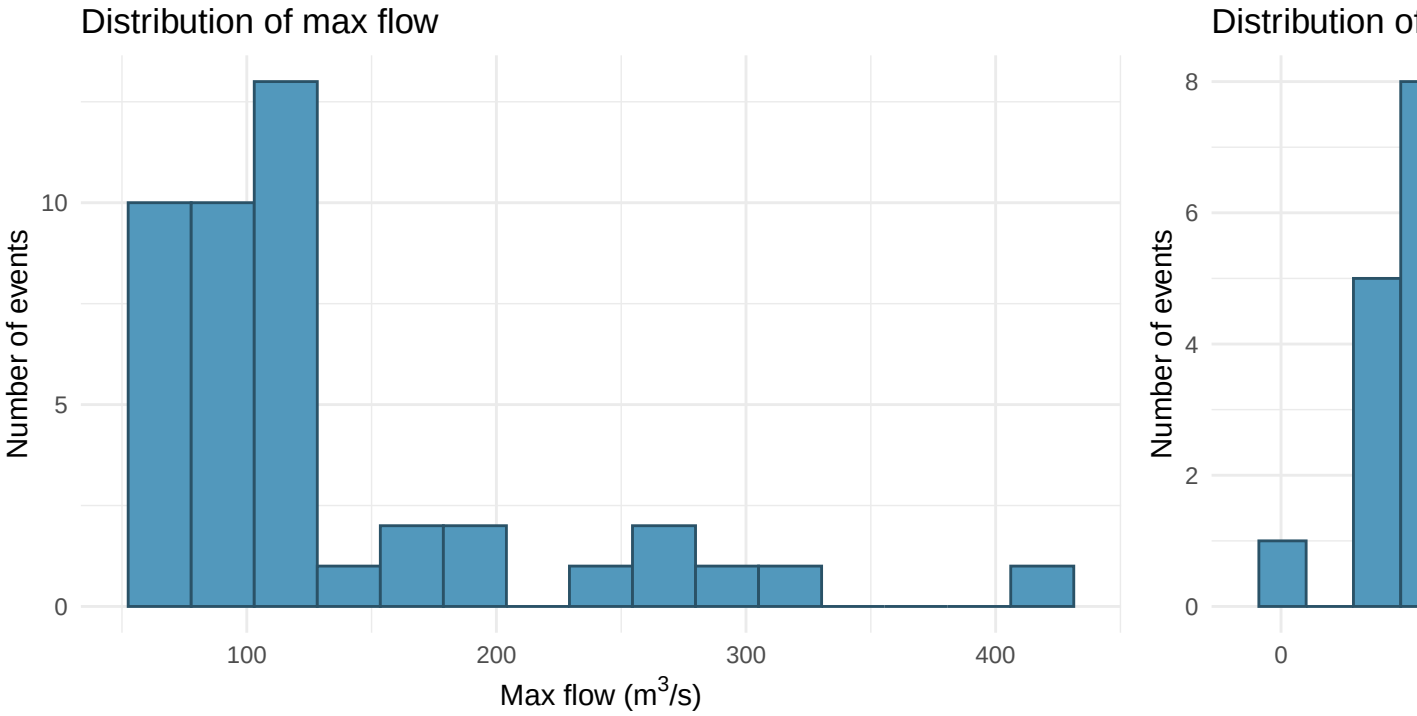


Figure 11: The Distrubtion of the two parameters

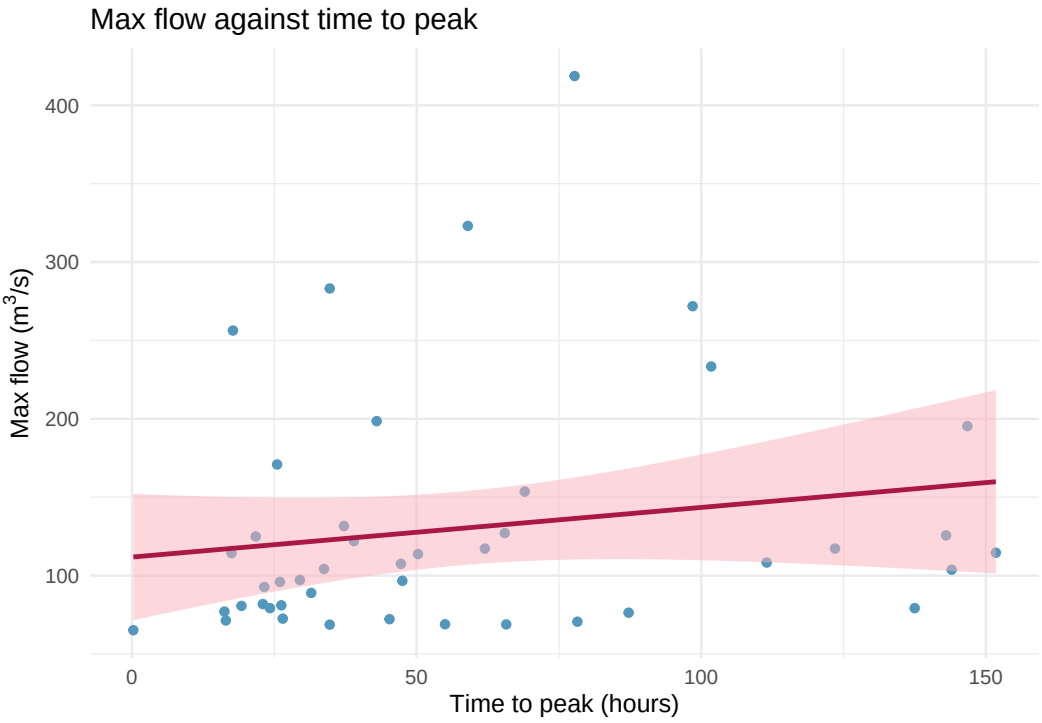


Figure 12: An analysis of uncertainty in the two parameters

To investigate their relationship, we calculated Spearman's rank correlation coefficient, obtaining $\rho = 0.287$ ($p = 0.059$). This indicates a weak positive association, but with insufficient evidence of a monotonic relationship at the 5% significance level. Thus, indicating larger flood peaks do not necessarily correspond to a longer time to peak.

Table 2: Spearman's Rank Correlation Between Peak Discharge and Time to Peak

Test	Rho	P-value	S statistic
Spearman's rank correlation	0.287	0.059	10111.57

Further seasonal analysis revealed more differences (Table 3). Autumn effects had a median maximum flow of 117.19m^3/s and time to peak of 59 hours, in comparison winter had 108.95m^3/s and 35.5 hours. The limited number in spring (6 events) and summer (1 event) restricts them from meaningful season analysis.

Table 3: Summary of Max Flow and Time to Peak for each Season

season	Events	Median max flow	Median time to peak (h)
Winter	18	108.951	35.500
Spring	6	80.812	30.625
Summer	1	88.889	31.500
Autumn	19	117.185	59.000

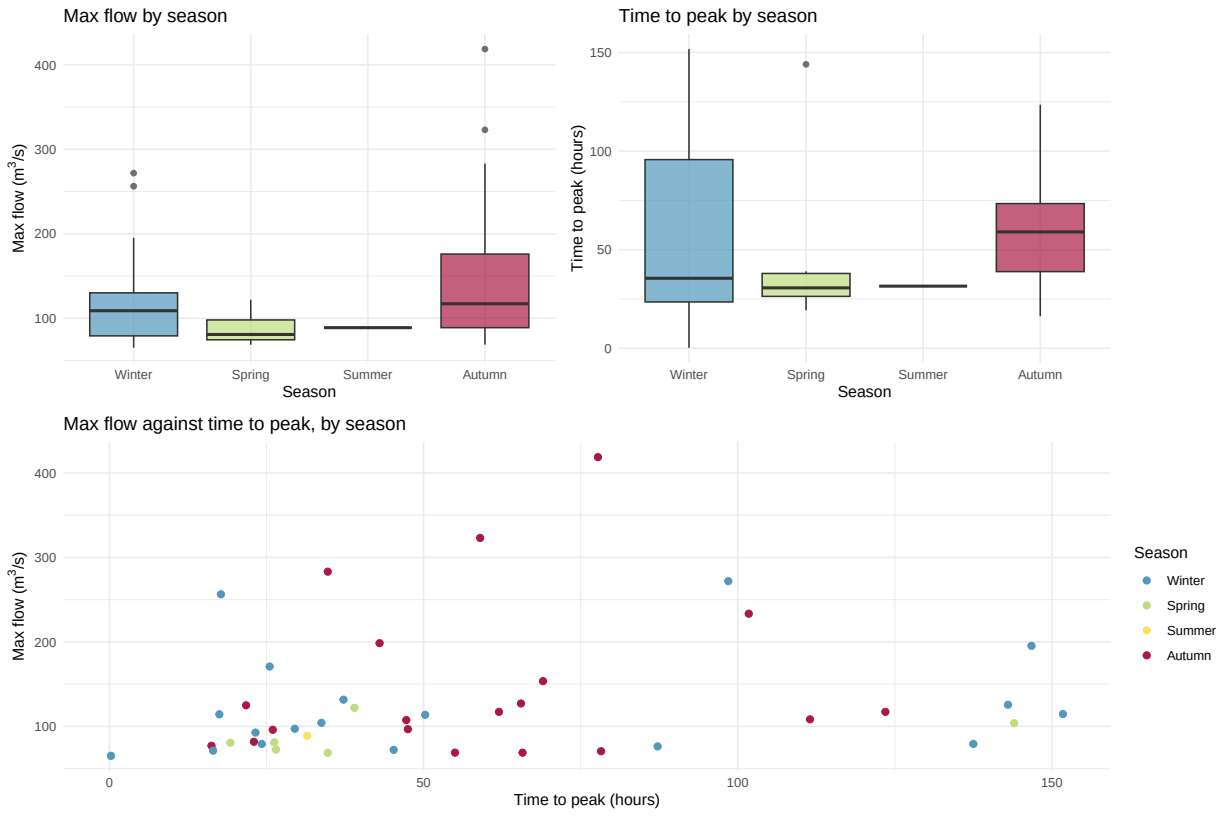


Figure 13: A comparision of the uncertanity in the parameters by season

Table 4: Final Uncertanity Table

threshold	events	Events per Year	Median Max Flow	Median Time to Peak (Hours)	Spearman Rho Corelation
55	57	5.7	81.792	43.000	0.066
60	46	4.6	100.434	44.125	0.184
65	44	4.4	105.834	44.125	0.287
70	39	3.9	108.308	43.000	0.270
75	37	3.7	114.256	47.250	0.220

We also assessed the sensitivity to threshold selection. Across the thresholds of $60 - 75 \mathrm{m}^3/\mathrm{s}$, median time to peak remained between 43 and 47.25 hours, while Spearman's correlation ranged from 0.184 to 0.287. This suggests that the observed variability in time to peak was not solely a consequence of the chosen threshold.

Overall, these findings support incorporating uncertainty in hydrograph timing as well as a maximum flow into flood inundation modelling, rather than assuming a fixed hydrograph shape.